Final Project Proposal
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BorrowSmart Debt Burden Pipeline

SECTION 1: ORGANIZATION & USE CASE

Our fictional organization is a national nonprofit, BorrowSmart that helps borrowers manage their federal student loan debt. The organization operates primarily online, allowing advisors to support borrowers across the United States.

BorrowSmart employs approximately 75 advisors and serves over 50,000 borrowers annually. BorrowSmart currently relies on manually reviewed federal reports and static summaries rather than an integrated analytical system.

BorrowSmart advisors rely mostly on national-level student loan statistics when helping borrowers choose repayment options. Advisors bring knowledge of state-specific repayment environments, but currently lack data tools to compare debt burden to local income levels. While national statistics provide general context, they do not reflect differences in local income levels. Someone living in a high-income state may be able to manage the same loan burden more easily than someone in a lower-income state. Without a way to compare debt levels to local income levels, advisors may unintentionally recommend repayment strategies that are not sustainable.

This project addresses the lack of regional context in repayment guidance by enabling advisors to evaluate how student loan burdens compare to income levels in a borrower's state. This helps identify which regions may be at higher risk of repayment stress or default.

This project proposes building a data pipeline that integrates federal student loan portfolio data with state-level median household income data to generate a state-level debt burden indicator.

The primary users of this system will be the advisors and program managers within BorrowSmart. Advisors will use the tool during borrower consultations, and program managers will use it to prioritize outreach efforts. This will enable BorrowSmart to identify where repayment support is most needed and better allocate advisory resources.

Success will be measured through:

- Creation of a quarterly state-level debt burden indicator
- Identification of the top 10 highest burden states
- Reduced reliance on national averages in advisory workflows, allowing the organization to better target repayment support and outreach efforts.

SECTION 2: DATA SOURCES & INITIAL ARCHITECTURE

This project will rely on publicly available government datasets.

Dataset 1: Federal Student Loan Portfolio by Location
Source: U.S. Department of Education, Office of Federal Student Aid
Type: XLS file
Update frequency: Quarterly
URL: https://studentaid.gov/data-center/student
Citation: Office of Federal Student Aid. (2025). Federal student loan portfolio by location [Data set]. U.S. Department of Education. https://studentaid.gov/data-center/student

Dataset 2: American Community Survey - Median Household Income
Source: U.S. Census Bureau
Type: CSV
Update frequency: Released annually (5-year rolling estimate) - Census adds a new year every year and then drops the oldest year
URL: https://data.census.gov
Citation: U.S. Census Bureau. (2024). American Community Survey 5-year estimates (2019–2023): Median household income (Table B19013). https://data.census.gov/table/ACSDT5Y2023.B19013

Dataset 3: Consumer Credit Trends - Student Loans
Source: Consumer Financial Protection Bureau (CFPB)
Type: CSV
Update frequency: Monthly
URL: https://www.consumerfinance.gov/data-research/consumer-credit-trends/student-loans/
Citation: Consumer Financial Protection Bureau. (2025). Consumer credit trends: Student loans [Data set]. https://www.consumerfinance.gov/data-research/consumer-credit-trends/student-loans/

Direct data file used:
https://files.consumerfinance.gov/data/consumer-credit-trends/student-loans/map_data_STU.csv

How the datasets connect:
All three datasets report at the state level, so we join them using state FIPS codes as the

common key. FIPS codes are the standardized numeric identifiers the government uses for each state, and they prevent matching issues from inconsistent state name formatting.

Step 1: Join the FSA loan portfolio data with the ACS income data on state FIPS codes. This gives us total outstanding debt and borrower counts alongside median household income for each state. From there ,we calculate a debt-to-income ratio per state.

Step 2: Join the CFPB consumer credit trends data onto that same table, again by state. This adds state-level year-over-year change in student loan originations.

What the final table looks like:
Each row is a state. Columns include:

- Total outstanding federal loan balance (from FSA)
- Number of borrowers (from FSA)
- Average debt per borrower (from FSA)
- Median household income (from ACS)
- Debt-to-income ratio (calculated)
- Year-over-year change in new loan originations (from CFPB)

Time alignment note: FSA updates quarterly, ACS is released annually as a 5-year rolling estimate, and CFPB updates monthly. We would align on the most recent overlapping period. For example, Q3 2025 FSA data matched the 2020-2024 ACS estimate and late 2025 CFPB data.

Future integrations may include additional aggregated consumer credit datasets to account for broader household debt exposure beyond federal student loans.

Data ingestion will occur through manual downloads of public datasets that are then loaded into Python for batch processing, which are standardized and transformed into a consistent format to ensure metadata standardization. Ingestion follows a pull pattern, as we retrieve data from public sources on our own schedule rather than receiving pushed updates. If requirements evolve, we will evaluate API-based ingestion based on latency and automation needs.

Primary storage will likely be a relational database as both datasets are structured and share a common state-level key. If query demands expand, we will evaluate columnar storage based on performance needs.

Processing will involve batch processing, which is appropriate given the quarterly and annual update cycles of our source data; real-time streaming would add unnecessary overhead for this use case. Key transformation steps include:

- Loading source files directly into Python
- Standardizing state identifiers to ensure consistency across datasets
- Aggregating and merging debt and income data by state

- Computing debt burden ratios
- Standardizing financial units so totals are comparable across datasets

An analytical layer will store derived metrics, enabling advisors to access simplified summaries rather than raw data and serving as a lightweight analytical data warehouse.

SECTION 3: DATA QUALITY & GOVERNANCE

To ensure reliability, the project will focus on completeness, accuracy, consistency, and timeliness.

Validation checkpoints will include (all done in Python):

We will standardize state identifiers across datasets to ensure they use the same format before merging. We will scan the data for missing or suppressed values and flag any states where key fields, such as total debt or median income, are incomplete. We will also track the reporting period or release date for each dataset to ensure we use the most recent available versions before performing comparisons. To verify accuracy, we will cross-check row counts and totals against published summary statistics from each source agency where available.

These checks occur at ingestion, after standardization, and before final analytical outputs are generated.

- verifying state identifiers match across datasets
- checking for missing or suppressed values
- confirming update recency
- Verifying join completeness by confirming the expected number of states remain after merging (50 states, as DC is not consistently reported across all sources)

Processing will halt if core metrics such as total debt or median income are missing. Non-critical gaps will be flagged for review.

Governance will include:

- metadata documentation for dataset definitions, including a data dictionary that defines each metric, its source, calculation method, and update frequency
- role-based access to derived outputs
- tracking of transformations applied during processing

We will control access by sharing only summarized results with advisors instead of raw data files. All transformation steps will be logged to maintain data lineage from source download to final analytical output.

This project will focus only on federally held student loan data to ensure consistency across states. State-specific aid programs and supplemental funding mechanisms are out of scope for

this phase, as they vary widely and are not consistently reported at the national level. Future versions of this system could incorporate state-level aid data where available.

At this stage, only aggregated public data is used, meaning there are no regulatory compliance requirements, and privacy risks remain minimal. However, the system may serve as a supporting decision tool in environments where federal student aid guidance capacity is limited.

SECTION 4: ETHICAL CONSIDERATIONS

Data Collection & Bias

One key consideration is representativeness. State-level averages may not reflect individual borrower circumstances. For example, a state's average debt burden could mask significant variation across income levels, age groups, or institution types within that state. There is also a risk that states with smaller populations or lower response rates may have less reliable estimates. Advisors will use the system as a guidance tool to support repayment decisions rather than as an automated decision rule. To mitigate bias, advisors will be trained to treat state-level metrics as context rather than deterministic guidance, and documentation will include explicit limitations about what the data can and cannot represent.

Transparency & Accountability

BorrowSmart prioritizes transparency in how data is used and transformed. Advisors should be able to explain how debt burden estimates are generated and how they influence recommendations. Documentation will explain how ratios are calculated and interpreted to avoid overgeneralization of state-level trends. BorrowSmart’s data team will be responsible for monitoring data accuracy and ethical use of the system outputs. A quarterly review process will assess whether the tool is being used appropriately and whether outputs remain aligned with the organization's advising standards.

Data Privacy & Protection

This project uses only aggregated, publicly available government data with no personally identifiable information (PII). Because the data is reported at the state level, individual borrowers cannot be identified from the system's outputs. However, if future iterations incorporate borrower-level data, BorrowSmart would implement data minimization principles, encryption, and access controls to protect sensitive information.